CONTROLLED RESEARCH AND AUTOMATION SPECIFICATION

Target-Price Revision ETF Alpha

Research, QuantConnect, and Bounded Autopilot Blueprint

Version 2.0 - August 29, 2026
A corrected replacement for the submitted v1 proposal. This document defines a separate target-price evidence family, a stock-first falsification path, an immutable QuantConnect handoff, and a promotion ladder that can evolve into bounded unattended trading only after prospective evidence, independent review, and explicit owner authorization.

Current state	What this document authorizes	What it does not authorize
Planning baseline only. No target-price implementation, production data, result, QC job, paper order, or live order exists.	Documentation, independent review, and a future TPR-0 contract-freeze milestone when separately scheduled.	Provider or outcome access, research looks, QC upload/backtest, broker action, paper/live deployment, capital allocation, or autonomous order authority.

Prepared for repository lane codex/strategy-target-price-revisions

Canonical repository folder: docs/Strategy Description

Source proposal SHA-256

53c549aef18aa1a63e6db8deb184bd654eb8ec637bb4ff3ae03f29abc4a2df0

Status legend: REQUIRED means a gate that must pass; PROHIBITED means a dangerous direction; DEFERRED means intentionally unavailable until a later authorization; DIAGNOSTIC means reportable but unable to rescue a failed primary test.

PART I

Executive Decision

The plan is feasible as a gated research program. Economic viability remains unknown, and production automation is a later promotion problem rather than a property of the backtest.

1. Decision and comparison

Decision: proceed as a separate lane, but not as submitted
The target-price hypothesis is technically buildable and economically distinct from rating revisions. The submitted plan is not safe to execute because its timing, as-of correction handling, independence, validity, coverage, statistical gates, and deployment language leave material degrees of freedom. Version 2.0 closes those openings and makes the first outcome access contingent on a reviewed TPR-0 specification.

Dimension	Submitted target-price plan	Active Analyst Revisions V2	This revised target-price plan
Research economics	Strong: continuous targets, target levels versus changes, price chasing, consensus state.	Strong for rating actions; target-price fields are explicitly a separate future family.	Retains target-specific strengths and defines one independent primary family.
Implementation contract	Incomplete: qualitative gates, conflicting coverage and timing, soft masking of hard invalidity.	Materially stronger typed contracts, refusals, evidence registries, timing and multiplicity controls.	Adopts the stronger safety rules and adds target-specific split, FX, horizon and consensus contracts.
Evidence today	None.	No accepted production signal or result; current contract candidate remains unaccepted.	None. The lane begins at documentation-only TPR-0.
Production readiness	Low; shadow-to-live language overreaches.	Research-only; no automatic deployment authority.	Explicit shadow, paper, canary and bounded unattended stages, each separately gated.

Calibrated assessment

- Research concept and target-specific design: about 8/10.
- Executable readiness of the submitted plan: about 3-5/10, depending on whether unresolved data semantics are treated as blockers.
- Expected readiness after this specification is independently accepted and its prerequisites are satisfied: about 8.5/10 as a research plan, not as proven alpha.
- Demonstrated alpha today: 0/10. No valid target-price outcome result exists.
- Engineering feasibility: yes. Economic viability: unknown. Data rights, point-in-time semantics, and permanent look authority are the critical path.

2. Why this is a separate family

Rating actions and target-price changes may share a provider row, but they are different hypotheses. A rating can remain unchanged while its target changes materially; a target can rise while the rating falls; and target levels can mechanically chase price. The target family therefore owns its own schema, preregistration, permanent look budget, result ledger, evidence epoch, and null-closure decision.

The rating engine is a mandatory control and, later, a candidate late-fusion input. It is never target-price ground truth. A valid null in either family cannot be rescued by the other family, ETF aggregation, a secondary horizon, or a combined model.

Governance correction
The repository's same-branch implementation/review exception names exactly three lanes and does not extend automatically. This fourth lane uses the generic topology: Codex implementation branch, independent Claude review branch, exact-snapshot review, Codex counter-review, and owner-controlled merge. Any future shared implementation requires a deliberate accepted synchronization, not copy-by-inference from an unaccepted analyst-lane head.

3. Core claims and non-claims

Class	Frozen statement
Primary claim	After strict point-in-time normalization, institution-deduped target-price revisions contain incremental short-horizon stock information after controlling for prior returns, rating actions, sector, and common catalysts.
Conditional ETF claim	Only if the stock claim passes once, a point-in-time ETF projection may preserve enough information after holdings lag, overlap, costs, and capacity to support an unlevered long-only allocation.
Automation claim	A deterministic QC consumer can reproduce an accepted offline decision packet and execute it safely under bounded authority. Automation is operational reproducibility, not evidence of alpha.
Non-claim	No historical result, expected return, win rate, confidence estimate, or production readiness is asserted by this document.
Null rule	A valid primary null closes the canonical target-price family. Secondary variants, ETF aggregation, and rating fusion may not reopen it.

PART II

Document Map and Frozen Vocabulary

The plan uses named states so missing evidence, invalid evidence, zero signal, and unavailable authority cannot be silently conflated.

4. Contents

5. Normative states

State	Meaning	Permitted behavior
VALID_ZERO	All hard contracts passed; the mathematically correct contribution is exactly zero.	Include as zero. Never treat as missing.
VALID_NONZERO	All hard contracts passed; a finite signed contribution exists.	Include under the frozen formula.
MISSING	Expected field or observation is absent and the absence may be ordinary.	Record the named missing reason; do not impute into the primary cell.
REFUSED	A hard validity contract failed or provenance is ambiguous.	Exclude from the primary score; retain raw evidence and refusal code.
INELIGIBLE	The security, event, ETF, or date is outside the frozen universe.	Exclude without treating it as a data defect.
STALE	A previously valid packet exceeded its freshness contract.	Block new or increasing exposure; preserve deterministic risk reduction.
UNAUTHORIZED	Required authority is absent even if code and data exist.	Do not access or run the gated surface.

Reliability is not confidence

Reliability is a preregistered measurement-quality scalar or status applied only after binary admission. It may reflect contributor breadth, active feature weight, holdings concentration, and measured source latency. It is not a probability of correctness, a calibrated expected return, or a license to override a refusal. The field name confidence is prohibited until a prospective calibration study passes its own frozen gate.

6. Canonical research parameters

TPR-0 must serialize every value below in a content-addressed preregistration before any outcome loader is made reachable. Values shown as FREEZE_AT_TPR0 are deliberate pre-outcome decisions that require the owner and independent reviewer; they may not be chosen after seeing returns.

Parameter	Canonical v2 setting or freeze rule
Primary event value	Price-scaled target change: (new_target - prior_target) / pre-event split-consistent stock price.
Event direction	Signed. Raises positive; cuts negative. Set/announce/initiation without a finite positive prior is not a revision.
Signal decay	Exponential, 20 exchange-session half-life, truncated after 80 sessions for the primary cell.
Scoring unit	At most one institution/security/session/catalyst contribution as of the decision cutoff.
Primary outcome horizon	20 exchange sessions, next-eligible-open to open after 20 sessions, net of frozen stock cost model.
Decision frequency	Weekly, first eligible exchange session of each week; holidays roll to the next session.
Primary comparison	Top-minus-bottom stock-score quintile, industry/sector controlled, on identical eligible observations.
Primary economic threshold	FREEZE_AT_TPR0 using a documented capacity/cost and power design; secondary results cannot replace it.
Independent sample floor	FREEZE_AT_TPR0 from event frequency, overlap, clustering and 80% power; no convenience-only universal count.
Multiplicity allocation	Zero until an external append-only authority assigns this fourth family an alpha/look budget and shared-holdout treatment.
Final shared holdout	Untouched. Its cutoff and reservation must be amended for four families before any target outcome access.

PART III

Point-in-Time Data and Event Contracts

Every accepted value must be reproducible from immutable evidence available before the decision cutoff. Unknown share basis, currency, identity, or timing refuses rather than receiving a soft haircut.

7. Source authority and zero-outcome pilot

The current provider documentation makes a structural audit plausible: fields are advertised for current and previous raw/adjusted targets, currency, target action, firm and analyst identifiers, event date/time, and last update. Documentation does not prove earliest public availability, complete correction vintages, fixed target horizon, split-restatement semantics, license rights for downstream QC processing, or historical completeness. Those are measured gates, not assumptions.

TPR-1 source audit - no returns permitted

- Pin provider, endpoint, API/schema version, request parameters, pagination behavior, timezone, and entitlement statement.
- Capture immutable raw response bytes, request/response metadata, fetch clock, page inventory, and SHA-256 manifest.
- Measure optional-field completeness, action vocabulary, currency distribution, positive/zero/nonfinite targets, adjusted/raw pair consistency, repeated identifiers, corrections, deletions, and late updates.
- Determine whether event time is demonstrably earliest public availability. If not, classify the row as date-only for canonical timing.
- Obtain explicit rights for local raw storage, normalized/derived processing, and the exact representation sent to QC Cloud or Object Store.
- Use only a small owner-authorized structural sample. Do not load future returns, rank candidate formulas, or spend a research look.

Fail-closed source gate
If the provider cannot support correction lineage, stable identities, share-basis reconciliation, target-horizon semantics, and permitted QC processing for the canonical sample, stop. A convenient endpoint is not sufficient evidence.

8. Four clocks and session eligibility

Clock	Definition	Canonical use
effective_at	When the analyst action purports to apply.	Descriptive only; never sufficient for tradability.
available_at	Earliest independently verified public availability in a named timezone.	Primary information-set clock.
ingested_at	When the controlled pipeline captured the version.	Latency and replay evidence; cannot move availability earlier.
version_available_at	When a correction or withdrawal version became available.	A correction changes only cutoffs at or after its own availability.

Eligibility rule

```
information_time = verified available_at, else conservative date-only time
research_cutoff  = frozen prior-session batch cutoff
eligible_open    = first exchange open strictly after max(information_time,
                                                          research_cutoff)

date-only row on event date D -> no earlier than the second exchange open after D
correction version V          -> usable only for decisions after V.version_available_at
```

Canonical v2 does not claim same-day premarket execution. The nightly packet uses a prior-session cutoff. A future pre-open extension would require authenticated intraday semantics, a separately reviewed pre-open builder, complete packet generation before the operational cutoff, and a new preregistration. It cannot be activated by changing a configuration flag.

Cutoff-safe reconciliation

For each decision cutoff, select the latest version available by that cutoff. Never use the final state later observed for the institution/security/day. Afternoon corrections cannot overwrite the morning information set. Every raw version remains immutable and linked by a stable event lineage key.

9. Canonical target event

Field group	Required content
Evidence identity	raw_object_hash, raw_row_locator, provider, endpoint_version, ingest_batch_id, correction_lineage_id.
Event identity	provider_event_id or deterministic collision-refusing key, firm_id, analyst_id when available, permanent_security_id, catalyst_id.
Clocks	effective_at, available_at with precision/evidence class, ingested_at, version_available_at, exchange_calendar.
Targets	prior_target_raw, new_target_raw, provider-adjusted counterparts, currency, share_basis, target_horizon, action.
Market basis	pre_event_price, price_timestamp, split_factor_asof, ADR_ratio_asof, FX_rate_asof, basis_evidence_hash.
Disposition	Exactly one of accepted, valid_zero, missing, refused, ineligible; named reason and policy version.

Action taxonomy

Observed state	Canonical disposition
Finite positive prior and new target; raise/cut or computable signed change	Eligible revision after every other hard gate passes.
Initiation/set/announce with no finite positive prior	Separate target-level diagnostic; never a revision.
Withdrawal/suspension/no-target state	Categorical event and active-roster change; target value is missing, never numeric zero.
Zero, negative, nonfinite, contradictory or currency-ambiguous target	REFUSED with exact reason.
Same target after valid reconciliation	VALID_ZERO; include as zero if the provider event is otherwise legitimate.

10. Share basis, currency, ADRs and horizon

- Preserve raw and vendor-adjusted targets. Audit the vendor's restatement policy across known splits before choosing the canonical pair.
- Never apply a split factor twice. The normalizer must prove that prior target, new target and pre-event price share one as-of share basis.
- Convert with point-in-time FX whose publication time precedes the decision cutoff. Spot, fixing and stale rules are versioned. Unknown currency refuses.

- ADRs require point-in-time depositary ratio, listing and underlying identity. Missing or ambiguous ratios refuse the primary event.
- Target horizon must be explicit and comparable between prior and new states. If the provider convention cannot be documented or audited, the canonical target-revision family remains blocked.
- Corporate actions, ticker changes, mergers, spin-offs, bankruptcies and delistings bind through permanent security identity, never a current-ticker join.

No soft repair
Currency, share basis, identity, availability and correction failures are binary validity failures. They are not multiplied by a low reliability score and passed onward.

11. Point-in-time stock universe and controls

TPR-0 freezes the complete eligible stock universe before returns: security type, primary listing, ordinary share/ADR policy, exchange, price/liquidity/capacity screens, sector/industry taxonomy, listing and delisting treatment, and minimum group size. Eligibility is evaluated point in time from content-addressed snapshots.

Control family	Purpose	Required point-in-time evidence
Prior returns	Separate target information from price chasing.	Adjusted price path available before event, with split/dividend policy.
Rating actions	Measure incremental target information on identical observations.	Accepted rating event under its own ontology; missingness retained.
Earnings/guidance	Cluster common catalysts and test announcement concentration.	Public event timestamp and stable issuer identity.
Sector/industry	Cross-sectional neutralization and clustering.	PIT classification with version/hash and no current-label backfill.
Liquidity/cost	Capacity and executable-return model.	PIT ADV, spread proxy, price and commission schedule.
Terminal outcomes	Avoid survivorship-biased returns.	Delisting, merger consideration, final cash and successor mapping.

PART IV

Canonical Stock Signal

The score is constructed only from cutoff-valid revisions, grouped first by institution and then discounted for common catalysts. Secondary target-level and residual models are diagnostics until separately promoted.

12. Primary event contribution

```
delta_i = (target_new_i - target_prior_i) / pre_event_price_i
age_i   = exchange sessions from eligible_open_i to decision_session
decay_i = 2 ** (-age_i / 20)
raw_i   = clip(delta_i, -CLIP_TPR0, +CLIP_TPR0) * decay_i

Hard prerequisites:
  finite positive prior target, new target and pre-event price
  common share basis, comparable target horizon, PIT currency conversion
  permanent security and institution identity
  event/correction version available before cutoff
```

CLIP_TPR0 is frozen from a zero-outcome structural distribution and documented source-error analysis. It is not retuned to improve returns. Log target change is a predeclared robustness diagnostic, not a rescue cell.

13. Institution and catalyst independence

Repeated rows do not create independent evidence. First aggregate all cutoff-valid revisions to at most one institution/security/session/catalyst contribution under a frozen reconciliation policy. Then compute stock strength and breadth across stable institutions. Separately compute effective catalyst count. The conservative breadth used by reliability is the minimum of institution-effective count and catalyst-effective count.

```
N_eff(values) = (sum(abs(values)) ** 2) / sum(values ** 2)

institution_strength = robust aggregate of one contribution per institution
N_inst = N_eff(institution contributions)
N_cat  = N_eff(catalyst-cluster contributions)
N_independent = min(N_inst, N_cat)
```

Raw event count, named analyst count and wall-clock burst size remain diagnostics. They cannot be described as independent contributor breadth. Unknown catalyst identity receives a conservative cluster policy frozen before outcomes.

14. Validity, reliability and normalization

Layer	Examples	Rule
Hard validity	Timing, version lineage, identity, share basis, target horizon, currency, finite values, PIT classification.	Any failure -> REFUSED. No score.
Signal strength	Signed price-scaled revision with time decay and robust institution aggregation.	May be zero or nonzero; never called confidence.
Reliability	Independent breadth, source precision, active evidence weight, concentration, measured latency.	Only after admission; monotone, bounded, preregistered and separately reported.
Normalization	PIT sector/industry robust centering and scaling.	Minimum group size and dispersion required; sparse or zero-MAD -> named unavailable state.

No invented variance

The primary normalizer uses a frozen robust center and scale only when the PIT group meets its minimum size and the measured dispersion is strictly positive. There is no epsilon denominator. If the group is sparse or zero-MAD, return a named refusal/unavailable state or use a separately frozen higher-level fallback whose eligibility is determined without outcomes.

15. Secondary and diagnostic features

Feature	Status in canonical v2	Anti-selection rule
Log target revision	Robustness diagnostic.	Cannot replace failed price-scaled primary.
Static target upside	Control/diagnostic only.	Must not be mislabeled as revision information.
Paired consensus change	Secondary, active-roster contract required.	Compare identical institution roster; withdrawals and stale targets explicit.
Consensus novelty	Secondary.	Expiry, roster, aggregation and correction rules frozen before outcomes.
Unexpected revision residual	Secondary research family unless promoted before outcomes.	Training-only fit; purge/embargo; no full-sample residualization.
Rating interaction/fusion	Deferred until both standalone families pass.	Identical rows, missingness retained, separate integration look and holdout.

PART V

Stock-First Falsification and Permanent Look

One exact primary study earns or closes the family. The plan does not build ETF machinery to manufacture a pass after stock-level evidence fails.

16. TPR-0 preregistration package

- Canonical provider/schema/version and source-rights decision.
- Complete event, identity, correction, timing, split, currency, ADR, horizon, catalyst and universe policies.
- One primary formula, decay, clip, decision frequency, outcome horizon, cost model, independent unit and estimator.
- Exact calendar partitions, purging/embargo, minimum power/sufficiency requirement, economic threshold and robustness constraints.
- Every primary, secondary and exploratory experiment ID with family denominator; external append-only look authority.
- Four-family multiplicity and untouched shared-holdout amendment, or explicit exclusion of TPR from that shared family.
- Stop rule: a valid primary null closes canonical TPR; invalid evidence is repaired without silently spending an extra look.
- Content-addressed reviewed-spec anchor. Outcome-loader imports remain physically unavailable before acceptance.

17. One primary stock cell

Element	Frozen design
Decision observation	Weekly PIT stock cross-section built from events eligible by the prior-session batch cutoff.
Signal	20-session-decayed, institution/catalyst-aware, price-scaled target revision; PIT robust industry/sector normalization.
Portfolio diagnostic	Equal-weight top quintile minus equal-weight bottom quintile on the complete eligible stock universe; no overlapping duplicate security positions.
Outcome	Next eligible open to the open 20 exchange sessions later, including terminal outcomes and frozen trading costs.
Controls	Prior returns, rating action, PIT sector/industry, earnings/guidance catalyst and liquidity/capacity.
Estimator	Predeclared cross-sectional/panel estimator with date and security dependence handled by block/cluster inference; exact implementation pinned at TPR-0.
Split discipline	Walk-forward only; training choices fixed without validation/test leakage; purge and embargo at least the outcome overlap.
Primary pass	Assigned family alpha, preregistered economic threshold, power/sufficiency floor, positive direction and frozen fold-level stability constraints all pass.
Primary fail	Valid non-pass closes the canonical family. Secondary cells and ETF aggregation cannot rescue it.

18. Multiplicity, trial accounting and dispositions

Every attempted cell is registered before outcomes, including horizons, half-lives, timing cohorts, catalyst cohorts, residual specifications, quintile/decile choices, portfolio alternatives, cost sensitivities, rating comparisons and ETF variants. The deflated-Sharpe trial count and family-wise/FDR ledger include abandoned and failed attempts. Code changes after a valid look create a new hypothesis family unless the preregistration explicitly classified the correction as an invalid-run repair.

Disposition	Definition	Next action
VALID_PASS	All evidentiary, statistical, economic and robustness gates pass.	TPR-5 ETF topology may be scheduled; no deployment authority.
VALID_NULL	Study is valid but any primary gate fails.	Close canonical target-price family; archive complete result.
INVALID_DATA	PIT, identity, correction, source or outcome contract failed.	Repair only the defect under independent review; rerun under same registered hypothesis if authorized.
INVALID_IMPLEMENTATION	Parity, leakage, estimator or code defect invalidated the run.	Correct, review and rerun; do not report performance as evidence.
INSUFFICIENT	Frozen independent-sample/power floor not met.	Wait for/add legitimately available data; do not loosen thresholds after looking.
UNAUTHORIZED	Look authority or source/outcome authority absent.	Do not run.

19. Required stock-study outputs

- Immutable input manifest, exact Git commit/tree, environment/dependency lock, calendar and configuration hashes.
- Complete accepted/refused/missing/ineligible counts by reason, provider version, date, sector and security type.
- No-cost and net-cost primary estimate, assigned-alpha result, confidence interval, cluster/bootstrap diagnostics and effective sample size.
- Predeclared fold table, exposure, turnover, capacity, concentration and terminal-outcome checks.
- Control and identical-row incremental comparison to rating actions without allowing rating availability to filter the target-only primary sample.
- A signed result record with one final disposition, permanent look receipt and explicit statement of whether ETF work is reachable.

PART VI

Point-in-Time ETF Projection

ETF construction is unreachable until the stock family earns a valid pass. Complete holdings books and raw weighted exposure prevent sparse coverage from being amplified into a false signal.

20. ETF reachability and universe

Hard stop/go gate
TPR-5 may not begin after a valid stock null. A plausible story about diversification, investor flows or packaging is not evidence that ETF aggregation can create information absent at the constituent level.

If reachable, the ETF universe is independently constructed from PIT listings, product type, inception, closure/liquidation, holdings availability, AUM/liquidity, leverage/inverse status and tradability. It is not seeded from today's surviving popular tickers. The canonical portfolio excludes leveraged, inverse, single-stock, commodity, crypto and non-equity products unless TPR-0 names a separate diagnostic universe.

Holdings availability

Each holdings snapshot carries source time, publication/availability time, effective date, capture time, full constituent book, residual/cash treatment, share-class policy and content hash. The strategy uses a fixed conservative availability lag proven by the source audit. Because QC constituent data can be delayed, a one-session assumption cannot be claimed without measurement; actual, one-session and five-session lags are predeclared sensitivity cohorts. Only the canonical audited lag drives the primary ETF result.

21. Three separate coverage concepts

Metric	Definition	Policy
mapped_holdings_weight	Absolute holdings weight bound to valid permanent security identities plus documented cash/residual states.	Hard gate: at least 99% for confirmatory ETF decisions; no caller-configurable bypass.
feature_observed_weight	Mapped weight with an admitted VALID_ZERO or VALID_NONZERO stock score.	Reported and used in monotone reliability; missing/refused are not converted to zero.
active_signal_weight	Mapped weight with a currently nonzero decayed target contribution.	Descriptive intensity; never a mapping-completeness substitute.

These fields replace the submitted plan's conflicting 30%, 50%, 90%, 95% and 99% uses of coverage. Thresholds for soft feature reliability are frozen at TPR-0; the 99% identity mapping gate is hard and explicit.

22. Raw ETF exposure and reliability

```
ETF_raw_e,t = sum over complete PIT holdings j of (weight_e,j,t * stock_score_j,t)

No division by covered weight.
No renormalization of mapped holdings into 100% risky exposure.
Missing/refused stock evidence stays missing/refused and lowers reliability.

ETF_reliability = monotone frozen function of:
  feature_observed_weight, independent contributor breadth,
  ETF constituent effective-N, HHI/concentration,
  holdings age/latency, and mapping completeness above the hard gate.
```

Portfolio ranking uses the preregistered relationship between raw exposure and reliability. A low-coverage ETF cannot obtain an extreme score merely because its small observed slice is renormalized. Residual cash is valid and expected.

23. Peer groups, overlap and comparisons

- Peer taxonomy is point-in-time and content-addressed. The source, lag, similarity metric, threshold, clustering algorithm, minimum peer count, singleton rule and zero-dispersion behavior are frozen before ETF outcomes.
- Holdings-overlap clusters constrain duplicate economic exposure. Correlated ETFs do not create independent breadth.
- Compare direct-stock, industry-basket and ETF implementations on aligned information sets and execution clocks.
- Compare target-only, rating-only and combined models on identical eligible rows. Missing or refused inputs remain explicit; combined coverage cannot cherry-pick the intersection and claim superiority.
- A combined model is evaluated only after both standalone families pass and under a separately allocated integration look. It cannot rescue either family.

24. Portfolio and cost contract

Element	Canonical safety contract
Style	Long-only, unlevered, bounded number of ETFs, residual cash permitted. Exact K and caps freeze at TPR-0 before ETF outcomes.
Eligibility	Valid complete packet, hard mapping gate, PIT product/liquidity screens, reliable quote/tradability and no prohibited product type.
Allocation	Deterministic target weights with name, peer-cluster, sector and turnover constraints; no ML/LLM order authority.
Exits	Complete target map includes zero targets for disappeared/ineligible positions. Forced risk exits cannot be blocked by stale new-signal data.
Costs	Commission + half-spread + square-root impact on net position change using PIT ADV, plus auction gap/slippage and rejection/unfilled behavior.
Sensitivity	0, 5, 10 and 20 bps-per-side diagnostics plus the canonical measured model. A favorable sensitivity cannot replace failure under the primary model.
Capacity	Participation, ADV, order-size and concentration limits frozen before QC outcomes; fills modeled at auction-compatible prices.

PART VII

QuantConnect Research and Execution Architecture

Provider normalization stays outside LEAN. QuantConnect consumes only complete immutable decision packets, reproduces accepted portfolio logic, and refuses any packet whose evidence or authority is incomplete.

25. Architecture boundary

Layer	Responsibility	Prohibited behavior
Offline evidence builder	Provider capture, bitemporal normalization, identities, splits/FX, stock scores, holdings projection, eligibility and complete ETF targets.	No outcomes before authority; no mutable overwrite of an evidence packet.
Artifact registry	Immutable objects, complete manifests, schema/config/calendar hashes, evidence epoch, decision cutoff and publication receipt.	No unversioned latest.csv as authority; no partial universe export.
QC research consumer	Read packet, validate manifest, reproduce rank/portfolio/cost logic, emit parity evidence.	No vendor API calls; no hidden feature engineering; no live authority.
QC execution consumer	Pre-open validation, target diff, bounded orders, state machine, reconciliation, telemetry and kill switch.	No parameter search, ML/LLM decision, retry-before-reconcile, or authority escalation.

Complete packet rule

Each packet contains every ETF in the frozen eligible universe, including zero/no-signal/ineligible/refused states and forced exits. Exporting only top-ranked names is prohibited because omitted positions become ambiguous and stale holdings can persist.

26. Manifest and decision packet

```
manifest:
  strategy_id, policy_version, evidence_epoch
  decision_session, information_cutoff, generated_at, expires_at
  git_commit, git_tree, environment_lock_hash
  schema_hash, config_hash, calendar_hash
  provider_snapshot_hashes, security_master_hash
  split_fx_basis_hash, catalyst_snapshot_hash
  holdings_snapshot_hashes, peer_taxonomy_hash
  cost_policy_hash, universe_hash, complete_row_count
  prior_manifest_hash, publication_receipt

row:
  permanent_etf_id, symbol_asof, state, raw_score, reliability
  mapped_holdings_weight, feature_observed_weight, active_signal_weight
  constituent_effective_n, hhi, target_weight, forced_exit_reason
  row_hash, decision_reason_codes
```

QC verifies all hashes, row count, completeness, monotone timestamps, expected predecessor, policy version and evidence epoch before computing any order. A packet is either atomically admitted or refused; partial ingestion is not a degraded mode.

27. Deterministic rebalance clock

Time	Action	Failure behavior
Prior session after close	Freeze source cutoff; build, validate, sign/hash and publish next eligible rebalance packet.	No packet publication on any incomplete input or failed contract.
Rebalance day 09:15 ET	QC pre-open revalidates freshness, calendar, holdings, positions, open orders, buying power and risk state.	Block additions; prepare allowable risk reductions; alert.
By 09:25 ET	Create idempotent MOO intents with opening-gap buying-power buffer and deterministic order tags.	If the venue/platform cutoff is earlier, the frozen schedule moves earlier; never submit near or after the accepted cutoff.
Auction and open	Observe acknowledgements/fills/rejections; do not infer fill from submission.	Ambiguous state -> query/reconcile before any retry.
Post-open	Reconcile broker/QC positions, cash, fills, target deltas and packet lineage; publish signed operational record.	Unresolved mismatch trips kill switch and prevents new exposure.
End of day	Health/SLO summary, risk/cost drift, stale-age and next-packet readiness.	Critical breach pages operator and keeps strategy fail-closed.

QuantConnect documentation currently requires MOO submission at least two minutes before the open and notes that auction orders can remain unfilled. The operational schedule therefore uses a larger buffer and freezes the exact tested cutoff before paper promotion.

28. Order idempotency and reconciliation

```
intent_key = hash(strategy_id, evidence_epoch, decision_session,
                  rebalance_id, permanent_security_id, target_version)

PREPARED -> SUBMITTED -> ACKNOWLEDGED -> PARTIALLY_FILLED
         -> FILLED | CANCELED | REJECTED -> RECONCILED

Restart rule:
  load durable intents and broker/QC state
  reconcile every ambiguous/submitted intent
  retry only if the prior intent is proven terminal and retry policy permits
```

- Order tags carry the intent key, packet hash and policy version. Duplicate intent keys are rejected.
- Target differences use reconciled positions and open orders, never an assumed local state.
- Opening-gap buying-power buffers and residual cash prevent prior-close sizing from exhausting buying power at the auction.
- Rejected or unfilled orders receive bounded deterministic handling. There is no unbounded chase loop.
- Risk-reducing exits are a separate authority class from new exposure. Stale signal data can block additions without stranding known risk.

29. QC parity gates

Gate	Acceptance evidence
Packet parity	Offline and QC decode identical rows, states, scores, reliabilities and target weights from the same manifest.
Clock parity	Research and QC select identical eligible events/sessions under DST, holidays, early closes and date-only rows.
Portfolio parity	Complete target maps, constraints, residual cash, exits and turnover match within frozen Decimal/tolerance policy.
Cost parity	Commission, spread, impact and auction assumptions use the same versioned inputs; deviations are explained and bounded.
Failure parity	Corrupt, partial, stale, future-dated, rollback, wrong-epoch and wrong-predecessor packets all refuse in both environments.
No authority crossing	Research backtest code cannot access broker credentials or submit orders; execution code cannot search parameters or outcomes.

PART VIII

Promotion to Bounded QuantConnect Autopilot

Autopilot means deterministic unattended operation inside explicit risk and authority limits. It does not mean unlimited capital, self-modifying strategy logic, or automatic promotion from a backtest.

30. Promotion ladder

Stage	Orders	Required evidence and decision
TPR-7 QC research parity	None	Accepted stock and ETF evidence, immutable packets, QC backtest parity, independent review. Research-only.
TPR-8 lane dossier	None	Complete result, limitations, costs/capacity, null/pass disposition, code/evidence hashes and unresolved risks. Shared final holdout remains closed.
TPR-9 live shadow	None	Real-time packet generation, clock/latency/quality telemetry, simulated intents and reconciliation drills. Separate owner scheduling.
TPR-10 QC paper autopilot	Paper only	Prospective sufficiency plan, frozen config, idempotency, restart and ambiguous-order drills, SLOs, alerts, kill switch, runbook and independent acceptance.
TPR-11 restricted live canary	Live, tiny fixed capital	Explicit owner authorization, approved broker/account/symbol/capital/loss limits, zero unresolved critical incidents, paper evidence sufficiency and tested rollback.
TPR-12 bounded unattended live	Live within fixed envelope	Prospective canary evidence, operational SLOs, risk/cost drift gates, independent promotion dossier and new explicit owner authorization.

No automatic arrows
A historical pass does not authorize shadow. Shadow does not authorize paper. Paper does not authorize live. Canary does not authorize unattended operation or capital expansion. Each transition is a separately reviewed owner decision bound to an exact snapshot and evidence dossier.

31. Paper evidence and operational sufficiency

Before TPR-10 starts, the paper protocol freezes a prospective evidence horizon based on rebalance frequency, independent observations, expected turnover and the effect/cost precision needed. Calendar duration alone is not sufficient. The protocol also defines an operational minimum window, eligible-session count, permitted resets, incident severity, and how rejected/unfilled orders affect sufficiency.

Evidence family	Examples of frozen promotion gates
Data/decision	100% manifest validation; no unknown version or rollback; bounded packet latency; complete universe; named stale/refusal rates within gate.
Parity	Paper target and simulated research target match; unexplained weight/order differences are zero.
Execution	Intent uniqueness 100%; all ambiguous orders reconciled before retry; fills/rejections and opening gaps within frozen policy.
Risk	Gross/net/name/cluster/cash limits never exceeded; drawdown and loss-budget gates; forced-exit path tested.
Operations	Restart recovery, stale packet, corrupt packet, broker disconnect, partial fill, calendar anomaly and kill-switch drills pass.
Economics	Prospective slippage, turnover and net performance remain within the preregistered promotion band; no favorable hindsight reset.

32. Risk envelope and kill switch

The risk envelope is explicit configuration approved at each live stage: account, symbols, product types, maximum capital, gross exposure, name/sector/peer-cluster concentration, turnover, order notional, daily loss, drawdown, stale age, data rejection, broker divergence and retry counts. Values are not inferred from backtest volatility and cannot expand automatically.

Immediate stop-new-risk triggers

- Missing, partial, corrupt, future-dated, wrong-epoch, wrong-predecessor or expired decision packet.
- Clock/calendar mismatch, unrecognized policy/schema version, evidence rollback, or non-monotone manifest chain.
- Position, cash, buying-power or open-order mismatch after bounded reconciliation.
- Duplicate/ambiguous intent, repeated reject, unbounded auction slippage, missing acknowledgement or broker/QC disconnect.
- Gross/name/cluster/turnover/loss/drawdown/capital breach, or a metric needed to prove a limit is unavailable.
- Telemetry/alerting failure at a stage where monitoring is a promotion requirement.

The kill switch cancels pending additions where safe, blocks new exposure, preserves evidence, alerts the operator and follows a separately tested risk-reduction/flattening policy. It never improvises. Restart requires a signed recovery checklist and full broker reconciliation.

33. Monitoring, drift and rollback

Surface	Monitored signals	Response
Source	Latency, optional fields, action vocabulary, corrections, currency/horizon/share-basis drift.	Refuse affected rows; escalate schema/source change; new policy version before reuse.
Signal	Coverage states, breadth, concentration, score distribution, sector availability and stale decay.	Block promotion or additions when gate fails; no automatic retraining.
Portfolio	Turnover, overlap, cash, exposures, capacity, target churn and forced exits.	Apply frozen caps; investigate sustained drift; no silent parameter change.
Execution	Gap, spread, impact, fill ratio, reject/cancel, time-to-ack and reconciliation mismatch.	Stop new risk at threshold; reconcile; operator decision.
Performance	Net return, drawdown, beta/factor drift and realized versus expected cost bands.	Promotion/continuation gate only; no self-tuning or automatic capital increase.
Audit	Manifest continuity, intent uniqueness, policy version, authorization receipt and incident closure.	Fail closed on missing audit evidence; rollback to last accepted immutable release.

Rollback means returning code/config to an exact accepted immutable release while retaining current broker state and reconciling positions under the approved recovery plan. It does not mean replaying old target packets or pretending current positions match the old release.

34. Human and model authority

Actor	Permitted role	Prohibited authority
Deterministic strategy code	Validate accepted packet, calculate bounded target difference, submit authorized orders, reconcile, enforce risk and emit evidence.	Cannot change hypothesis, parameters, risk envelope, capital or promotion stage.
ML model	None in canonical v2. A future residual model is research-only under separate preregistration.	No direct or indirect order authority; no online adaptation.
LLM/agent	Explain records, draft review material, surface anomalies under audit.	No signal, target weight, order, retry, promotion, capital or kill-switch override authority.
Owner	Explicitly approve sources, looks, QC runs, paper/live stages, accounts, capital and risk envelopes.	Approval must bind exact artifacts; silence is not authorization.
Independent reviewer	Review exact snapshot, evidence, gates and defects; produce P0-P3 disposition.	Review is not owner authorization and cannot create trading authority.

PART IX

Implementation Program and Definition of Done

Each milestone is bounded, independently reviewable, and unable to reach the next authority surface by configuration alone.

35. Milestone ladder

ID	Milestone	Definition of done / authority ceiling
TPR-0	Coordination and frozen spec	Owner decisions, fourth-family multiplicity/holdout, exact source contract, policies, primary cell, permanent look, reviewed hash. Outcome loader remains unreachable.
TPR-1	Immutable target ingest	Strict Decimal parser, actions/corrections/IDs/clocks, complete dispositions, structural audit and rights evidence. No returns.
TPR-2	PIT prerequisites	Security/institution/catalyst masters, price/split/FX/ADR/horizon, controls, terminal outcomes and cost/ADV contracts. No alpha run.
TPR-3	Canonical stock score	Fixture-only event aggregation, decay, robust normalization, validity/reliability, diagnostics and dangerous-direction tests.
TPR-4	One-shot stock study	Exactly one authorized primary outcome look and final pass/null/invalid/insufficient disposition. Valid null closes family.
TPR-5	PIT ETF topology	Reachable only after TPR-4 pass. Complete holdings, reverse index, universe, mapping, peers, overlap and raw exposure.
TPR-6	ETF walk-forward	Frozen portfolio, direct/industry/ETF and rating-increment comparisons, costs, capacity, full result dossier. No QC/deployment authority.
TPR-7	QC research parity	Immutable custom/precomputed packet consumer, backtest parity, failure parity and accepted exact snapshot. No orders.
TPR-8	Lane dossier	Independent final lane review, limitations, evidence hashes, integration recommendation; shared final holdout stays untouched.
TPR-9	Live shadow	Real-time no-order operation, operational drills, telemetry and prospective protocol under separate scheduling.
TPR-10	QC paper autopilot	Paper orders only, prospective sufficiency, idempotency/reconciliation, runbook, kill switch and independent promotion dossier.
TPR-11	Restricted live canary	Explicit owner authorization; tiny fixed capital/risk envelope; reversible; no expansion by inference.
TPR-12	Bounded unattended live	New owner authorization after canary evidence; fixed envelope, SLOs, monitoring, rollback. Each later expansion is a new decision.

36. Repository-native implementation boundaries

- The lane begins from main commit 086b782e43a5ff889e71ec8e26334bb791ccac74, not the unaccepted Analyst Revisions candidate.
- Accepted analyst-lane infrastructure may be reused only after exact independent acceptance and a deliberate reviewed sync/extraction. Rating-bound schemas, score meanings and evidence epochs are never relabeled as target artifacts.
- New code and tests use a target-price-owned namespace and avoid frozen shared files. A true shared dependency stops for owner-directed coordination.
- The target PDF and implementation record are the lane authority. The Action Plan and Session Handoff receive only concise coordination/state references when owner direction changes the fourth-lane sequence.
- Generic review applies unless the owner explicitly extends the same-branch exception: Codex stable commit -> Claude separate review branch -> Codex counter-review -> owner-controlled merge.

- No push, provider access, outcome load, QC job, broker action, scheduled task or deployment is implied by creating the worktree or accepting this document.

37. Automated verification

Test family	Required dangerous-direction cases
Events	Zero/negative/nonfinite targets; missing prior; withdrawal; initiation; duplicate; late correction; same-day after-cutoff version; unknown action; collision.
Timing	DST, early close, holiday, date-only second-open rule, precise event after cutoff, correction availability, clock-zone ambiguity.
Basis	Split before/between/after targets, vendor-adjusted double-adjustment trap, FX missing/stale, ADR-ratio change, horizon mismatch.
Identity	Ticker reuse, class change, merger, spin-off, delisting, firm rename/succession, analyst move, ambiguous mapping.
Signal	One firm repeated rows, many analysts/one firm, many firms/one catalyst, zero MAD, sparse sector, decay boundary, clip boundary, valid zero.
ETF	Incomplete book, 98.99% mapping refusal, missing versus zero feature, no covered-weight amplification, stale holdings, dead ETF, overlap singleton.
Statistics	Look receipt absent, unregistered cell, shared-holdout leak, overlap without purge, training/test contamination, secondary rescue attempt.
QC packet	Missing row, partial upload, corrupt hash, future date, rollback, wrong epoch/predecessor, stale packet, complete forced exits.
Orders	Duplicate start, restart after submit, ambiguous acknowledgement, partial fill, reject, opening gap, buying-power breach, stale-addition block with exit allowed.
Authority	Research code cannot import broker surface; execution cannot import outcome/search surface; LLM/ML output cannot reach order construction.

Validation proportional to milestone

Each stable snapshot runs targeted unit/property/integration tests, dangerous-direction mutations, repository governance guards, compile/static checks, diff hygiene and secret-shape scans. Outcome milestones additionally prove input immutability, exact look receipt, deterministic replay and result-record completeness. QC/paper/live milestones add platform parity, fault injection, restart, reconciliation and rollback drills.

38. Risk register

Risk	Consequence	Primary mitigation / stop
Target changes mostly chase price	False alpha story; factor exposure.	Prior-return controls, training-only residual diagnostic, one stock-first test; valid null closes.
Provider time is not public time	Look-ahead bias.	Conservative date-only eligibility; precise timing unusable until independently audited.
Vendor restates old targets for splits	Double adjustment and artificial revisions.	Raw/adjusted reconciliation audit; common share-basis proof; refuse ambiguity.
Targets lack consistent horizon/currency	Incomparable values.	Hard horizon/currency/FX/ADR contracts; stop if provider cannot support them.
Correlated analysts/catalysts	Inflated evidence breadth.	Firm aggregation and min(institution-effective, catalyst-effective) breadth.
Sparse ETF coverage	Extreme amplified scores.	99% identity mapping gate, raw weighted exposure, explicit feature weight and reliability.
Researcher degrees of freedom	Selection bias and invalid significance.	One primary cell, permanent look receipt, complete trial ledger, immutable holdout.
QC/live parity drift	Orders differ from accepted research.	Immutable complete packets, exact parity gates, fail closed on version/hash mismatch.
Ambiguous/repeated orders	Duplicate exposure or uncontrolled risk.	Durable idempotent intents, reconcile-before-retry, bounded state machine and kill switch.
Autopilot overreach	Unreviewed capital or strategy changes.	Separate stage authorization, fixed envelope, no self-tuning, new approval for every expansion.

39. Current evidence and authority matrix

Surface	Evidence today	Authority today
Revised planning document	Created as a documentation candidate; independent review pending.	Documentation only.
Provider entitlement/rights	Not established by this task.	Zero access.
Authenticated target events	0	Zero access.
Accepted target scores	0	Unauthorized/unimplemented.
Permanent target research looks	0	No receipt/authority.
Stock/ETF results	0	Unauthorized.
ETF topology/portfolio	0 production artifacts	Unimplemented; unreachable before stock pass.
QC projects/jobs/results	0 for TPR	No upload, compile, processing or backtest authority.
Shadow/paper/live orders	0	No broker/account/order/deployment authority.
Autopilot promotion	No prospective evidence	Deferred; separate future owner decisions.

40. Exact next step

Next: independent documentation review, then TPR-0 only
Commit this PDF and its implementation record as a stable documentation-only snapshot. Claude reviews that exact commit on a separate review branch under the generic workflow. After Codex counter-review and owner acceptance/scheduling, TPR-0 may freeze coordination, source, preregistration, holdout and look-authority decisions. Do not begin provider or outcome access while any of those gates is unresolved.

PART A

Appendices

Reference schemas and external facts support implementation but do not weaken the normative gates above.

A1. Minimal canonical event schema

```
TargetRevisionEventV2:
  evidence: raw_hash, row_locator, endpoint_version, batch_id
  identity: event_lineage_id, version_id, firm_id, analyst_id,
            permanent_security_id, catalyst_id
  clocks: effective_at, available_at, availability_precision,
          ingested_at, version_available_at, eligible_open
  target: prior_raw, new_raw, prior_adjusted, new_adjusted,
          action, currency, share_basis, target_horizon
  basis: pre_event_price, split_factor_asof, fx_rate_asof,
         adr_ratio_asof, basis_evidence_hash
  state: VALID_ZERO | VALID_NONZERO | MISSING | REFUSED | INELIGIBLE
  reason_codes: exhaustive list
  policy_version, schema_hash, row_hash
```

A2. Example operational decision record

```
decision_session: 2028-04-03
information_cutoff: 2028-03-31T18:00:00-04:00
evidence_epoch: tpr-paper-001
manifest_hash: <sha256>
policy_version: tpr-v2-paper-frozen
packet_state: ADMITTED
universe_rows_expected: 1847
universe_rows_observed: 1847
additions_allowed: true
risk_reductions_allowed: true

intent:
  intent_key: <sha256>
  permanent_security_id: <id>
  symbol_asof: <ticker>
  prior_weight: 0.00
  target_weight: 0.18
  order_type: MOO
  state: RECONCILED
  packet_hash: <sha256>
```

Illustrative values show the audit shape only. They are not a recommendation, result, target allocation, trading instruction, or live configuration.

A3. Provenance

Artifact	Pinned identity / role
Submitted target-price plan	TARGET_PRICE_REVISION_ETF_ALPHA_RESEARCH_QC_PLAN.pdf; 35 physical pages; SHA-256 53c549aef18aa1a63e6db8deb184bd654eb8ec637bb4ff3ae03f29abc4a2df0. Proposal content only.
Active analyst blueprint	ANALYST_REVISIONS_ETF_STRATEGY_BLUEPRINT_V2_EN.pdf; 64 pages; SHA-256 eae7b9954aaf94212108505c52e31a558facd744967fd2526040d5147c616193. Comparison and reusable-safety reference, not target authority.
Target lane base	main commit 086b782e43a5ff889e71ec8e26334bb791ccac74; branch codex/strategy-target-price-revisions; sibling worktree trading_agent_TargetPriceRevision.
Analyst lane comparison snapshot	origin/codex/strategy-analyst-revisions-v2 at 56d6fe0eff32d00b1692b3b17a3838649eeba56b. Unaccepted ARV2-2 candidate; not a base or accepted dependency.

A4. Selected external references

- Massive, Benzinga Analyst Ratings REST documentation: https://massive.com/docs/rest/partners/benzinga/analyst-ratings
- Massive, partner data overview and package information: https://massive.com/docs/rest/partners/overview
- Benzinga, Analyst Ratings API product page: https://www.benzinga.com/apis/cloud-product/analyst-ratings-api/
- QuantConnect, US ETF Constituents documentation: https://www.quantconnect.com/docs/v2/writing-algorithms/datasets/quantconnect/us-etf-constituents
- QuantConnect, Market On Open orders: https://www.quantconnect.com/docs/v2/writing-algorithms/trading-and-orders/order-types/market-on-open-orders
- Brav and Lehavy (2003), An Empirical Analysis of Analysts' Target Prices: https://doi.org/10.1111/1540-6261.00593
- Jegadeesh, Kim, Krische and Lee (2004), Analyzing the Analysts: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.2004.00657.x
- Journal of Economics and Business (2024), target-price changes and subsequent returns: https://doi.org/10.1016/j.jeconbus.2024.106174

External documentation is time-sensitive and must be reverified at each source/QC contract freeze. Academic findings motivate hypotheses; they do not establish that this implementation has alpha.

A5. Version 2.0 correction ledger

ID	Correction
C01	Separated target-price and rating families, looks, evidence epochs and null decisions.
C02	Replaced final-state same-day reconciliation with cutoff-safe version selection.
C03	Removed canonical same-day premarket assumption; added conservative session eligibility.
C04	Aggregated by institution and discounted common-catalyst dependence.
C05	Separated hard validity from soft reliability; prohibited uncalibrated confidence.
C06	Made split, adjusted-target, FX, ADR and target-horizon ambiguity fail closed.
C07	Added PIT stock universe, terminal outcomes and controls.
C08	Removed epsilon-based fake dispersion; sparse/zero-MAD states refuse.
C09	Separated identity mapping, observed features and active signals; fixed 99% mapping gate.
C10	Made peer taxonomy and holdings lag content-addressed pre-outcome contracts.
C11	Reduced confirmatory research to one primary cell and complete permanent trial accounting.
C12	Added exact disposition/stop rules; secondary/ETF cells cannot rescue a null.
C13	Expanded execution cost, capacity, MOO gap and rejected/unfilled modeling.
C14	Made QC consume complete immutable packets with no provider API or hidden feature build.
C15	Added exits, stale-addition/risk-reduction separation, idempotency and reconcile-before-retry.
C16	Replaced automatic deployment arrow with independently authorized shadow, paper, canary and bounded unattended stages.
C17	Added monitoring, SLOs, kill switch, restart recovery, rollback and fixed capital/risk envelopes.
C18	Pinned generic separate-review topology for the fourth lane unless owner explicitly changes it.

End state of this document
The plan is executable as a sequence of controlled milestones. The strategy is not yet proven, implemented, or authorized to trade. The first honest action is an independent review of this documentation snapshot, followed only by TPR-0 if the owner schedules it.